

CY 701: Advances Physical Methods in Chemistry Theory and Applications

Course Code	CY 701
Course Name	Advances Physical Methods in Chemistry Theory and Applications
L-T-P-C	3-0-0-3
Prerequisites	Consent of the faculty member
Intended for	Ph.D.
Distribution	Elective
Approval	1st Adhoc Senate (Apr 2011)

Course Contents

- **Quantum Chemistry and molecular symmetry**
Schrödinger wave equation, Born Oppenheimer Approximation and electronic wave function, particle in a box, potential well, potential barrier and tunneling. Definition of a group and basic theorems, molecular symmetry groups and classes, symmetry and normal modes of vibrations, determining the symmetry of molecular motions, symmetry and selection rules for allowed transitions among rotational, vibrational level, group theory and molecular electronic transitions. [8 Lectures]
- **Interaction of light with matter**
Transition moments and transition probabilities, Einstein's coefficients, oscillator strength, Beer-Lambert law, polarizabilities, Frank Condon Principles, Fluorescence and absorption spectroscopy, Quantum yield and its measurements. Donor acceptor complexes. Fluorescence quenching (static and dynamic). Stern Volmer analysis. Timescale of molecular processes in solution. Steady state and time-resolved fluorescence. Fluorescence anisotropy. Biochemical fluorophores. New fluorescence technologies: Multiphoton Excitation, Fluorescence correlation Spectroscopy, Single molecule detection. [10 Lectures]
- **Rotational and Vibrational Spectroscopy**
Moment of inertia, The Rotational energy levels, Rotational spectra of diatomic molecules, Rotational Raman spectra, Molecular vibrations, Selection rules, vibrational spectra of diatomic molecules, simple harmonic oscillator and rigid rotor model, anharmonic corrections, Vibrational-Rotational Spectra, vibrational Raman spectra of diatomic molecules. Infrared absorption spectra of polyatomic molecules, symmetric and asymmetric top molecules, normal modes of vibration and their classification by group theory, coupling between rotational and vibrational degrees of freedom. [7 Lectures]
- **Nuclear Magnetic Resonance Spectroscopy**
Basic Principles, Chemical shifts, Spin-spin interactions. Application of ^1H and ^{13}C NMR spectroscopy including COSY, NOESY in the structure determination of bioorganic compounds. Application in conformational analysis. Solid state NMR. Instrumental aspects. Multinuclear NMR of various inorganic and organometallic compounds. Instrumental aspects. [10 Lectures]
- **Mass spectrometry**
Basic concepts. Fragmentation and rearrangements (including McLafferty rearrangement) of different classes of organic molecules. Isotope effect. ESIMS, MALDI-TOF techniques. [5 Lectures]

Text Books

1. Donald A. McQuarrie and John D. Simon, **Physical Chemistry: A Molecular Approach**, University science books.
2. Peter Atkins and Julio de Paula, **Physical Chemistry**, OUP.
3. K Veera Reddy, **Symmetry and Spectroscopy of molecules** New Age International publishers
4. Joseph R. Lackowicz, **Principles of fluorescence spectroscopy**, Springer.
5. J. M. Hollas, **Modern Spectroscopy**, 4th Edition, John Wiley, 2004.
6. Edmond de Hoffmann and Vincent Stroobant, **Mass Spectrometry: Principles and applications**, 3rd edition John Wiley.